

Coding & Solution

Date	7 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Abhi-4363/smart_agriculture
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask
Key Features Implemented	Crop recommendation using machine learning, user input validation, soil parameter analysis, real-time prediction, responsive web interface
Pending / Incomplete Features	Multilingual support
Setup / Run Instructions	1. Install Python 3.10+ and required libraries. 2. Place the trained model.pkl file in the project folder. 3. Run app.py using Flask. 4. Open the application in a web browser and enter soil parameters to receive crop recommendation.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The application follows a modular architecture with separate frontend, backend, and machine learning components.
- The machine learning model was trained in Jupyter Notebook and saved as a Pickle (.pkl) file.
- The Flask backend loads the trained model to generate real-time crop recommendations.
- The application is deployed online and can be accessed through the hosted link.
- The project is designed to support future enhancements such as weather integration and multilingual support.